

Week 9 — Activity: Thermal Diffusivity, Thermal Length, and Timescales

Geophysics 3A: Potential Fields & Geothermics

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Purpose

To understand transient heat flow in terms of:

- how far thermal signals propagate in a given time,
- how long it takes thermal changes to affect a given depth,
- and how the *thermal length scale* controls signal shape and timing.

Background

Thermal diffusivity is defined as

$$\kappa = \frac{k}{\rho c_p},$$

where k is thermal conductivity, ρ is density, and c_p is specific heat.

Material	Thermal Conductivity k (W m ⁻¹ K ⁻¹)	Thermal Diffusivity κ (m ² s ⁻¹)	Thermal Diffusivity κ (km ² My ⁻¹)
Ice (glacial / ice sheet average)	1.6–2.0	$(0.7\text{--}0.9) \times 10^{-6}$	22–28
Clay-rich sediments	0.6–1.5	$(0.2\text{--}0.5) \times 10^{-6}$	6–16
Granite	2.5–3.5	$(0.8\text{--}1.2) \times 10^{-6}$	25–38
Basalt / Gabbro	1.7–2.2	$(0.6\text{--}1.0) \times 10^{-6}$	19–32
Peridotite (upper mantle)	3.3–4.5	$(1.0\text{--}1.5) \times 10^{-6}$	32–47

Values are representative ranges. Thermal conductivity and diffusivity vary with temperature, porosity, mineralogy, fluid content, and fabric. Diffusivities are calculated using $\kappa = k/(\rho c_p)$ and rounded for scaling arguments rather than precise modeling.

For diffusion-dominated heat transfer, the characteristic depth over which temperature changes propagate scales as

$$z \sim 2\sqrt{\kappa t}.$$

This thermal length sets both:

- the depth to which temperature changes have propagated after time t ,
- the timescale required for a thermal perturbation at depth to influence shallower levels.

Error Function Solution

For instantaneous heating or cooling at a boundary, the temperature evolution is given by the error function solution:

$$T(z, t) = T_s + (T_b - T_s) \operatorname{erf}\left(\frac{z}{2\sqrt{\kappa t}}\right).$$

Important: The argument of the error function,

$$\eta = \frac{z}{2\sqrt{\kappa t}},$$

controls the *shape and timing* of the thermal signal.

Tasks

1. For timescales of 10^3 , 10^5 , and 10^7 years, describe how the depth of thermal influence would differ between rock and ice.
2. Explain why borehole paleothermometry in rock requires much greater depths to recover long-term climate signals than ice-core records.

Tasks

A. Distance Given Time

1. For a fixed time t , describe how temperature varies with depth z as the value of η increases.
2. Explain why temperature changes are largest for $\eta \ll 1$ and become negligible for $\eta \gg 1$.
3. Using the thermal length ℓ_T , estimate the depth to which a surface temperature change has propagated after a given time.

B. Time Given Distance

1. Rearranging the thermal length relation,

$$t \sim \frac{z^2}{\kappa},$$

explain what sets the timescale for heat to travel a distance z .

2. Consider a plume impinging on the base of the lithosphere. If the lithosphere is x km thick, what controls how long it takes for this thermal perturbation to influence surface temperatures?
3. Following delamination of a dense crustal root, heat is supplied from below. Explain why crustal heating is delayed rather than instantaneous.

C. Interpreting “Arrival” of Heat

1. Does heat arrive at a specific time, or does it diffuse gradually? Use the error function shape to justify your answer.
2. For $\eta \approx 1$, what fraction of the total temperature change has occurred? What does this imply about defining a thermal response time?
3. Discuss why the concept of a sharp thermal front is inappropriate for diffusion-dominated heat transport.

Geological Interpretation

- Thick lithosphere acts as a strong thermal filter, delaying surface response to basal heating.
- Thermal signals generated at depth are attenuated and spread in time as they propagate upward.

- The thermal length scale controls which geological processes can be observed at the surface and on what timescales.

Key Insight

In diffusion problems, time and distance are interchangeable through the thermal length $\ell_T = 2\sqrt{\kappa t}$. Heat does not “arrive” at a depth at a single moment; instead, the error function describes a gradual, scale-dependent thermal response controlled by diffusivity, distance, and boundary conditions.